ITEP 414 / WEEK 02 PORTFOLIO PROJECT

ENTERPRISE INFRASTRUCTURE PLAN

ABC Startup Solutions | Proposed greenfield design

20	04	02	08
EMPLOYEES	DEPARTMENTS	WAN LINKS	REPORT PARTS

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Executive decision brief

ABC Startup Solutions is a 20-person software development startup occupying one office floor. It currently has no user devices, server, office network, internet service or security policy. This report proposes a controlled, right-sized environment that enables daily operations while preserving room to grow. All specifications and quantities are planning recommendations, not evidence of an actual purchase or deployment.

Recommended design at a glance

Capacity	Proposed approach
Endpoints	9 desktops (4 HR, 5 Finance) and 11 laptops (5 IT, 6 Sales); 22 external monitors.
Compute / storage	One Ubuntu Server host; one 4-bay NAS; two encrypted, rotating external backup drives.
Connectivity	Primary and backup business internet circuits into a dual-WAN router; dedicated firewall.
Local network	48-port managed PoE+ switch, 2 Wi-Fi 6 access points, 32 planned data outlets and 8 VLANs.
Governance	Microsoft 365 Business Premium, managed endpoints, multifactor authentication, backup/restore tests.

Decision required: Approve a site survey, request two independent ISP quotes, confirm software licensing and endpoint purchase specifications, and authorize a phased implementation. Vendor pricing and exact device models remain subject to procurement.

Planning assumptions and boundaries

Single floor; approximately 250-350 sq m assumed for preliminary Wi-Fi planning; 20 employee accounts; normal office hours; cloud-based email and collaboration; limited local development/staging requirements. The office is fictional. No floor plan, budget ceiling, actual supplier offers, compliance assessment, service-level commitment or physical deployment data was supplied. Quantities must be confirmed by measurement and management approval.

01 / Company profile

Attribute	Description
Company name	ABC Startup Solutions
Nature of business	Custom software development, business application support and digital solutions for small-to-medium clients.
Vision	To be a trusted technology partner delivering dependable and accessible software solutions to growing businesses.

Attribute	Description
Mission	To design and support secure, practical software solutions through skilled people, consistent service and responsible technology practices.
Office location	Suite 5, Innovation Park Center, Santa Rosa City, Laguna, Philippines - **fictional scenario location**.
Operating model	One office floor, mixed desk-based and mobile work, cloud productivity tools and controlled local services.

Organizational structure and employee distribution

Reporting line: **Executive sponsor** (existing management responsibility) -> **IT lead, HR lead, Finance lead, Sales lead** -> their respective team members. Department leads are assigned within the existing staffing allocation; the table below does **not** add extra employees. The five-person IT group includes junior system administration responsibilities.

Department	Headcount	Infrastructure dependency
Information Technology	5	Development laptops, Git, test VMs, server access and elevated management tools.
Human Resources	4	Secure desktop workstations, confidential personnel files and shared printing.
Finance	5	Stable desktops, restricted financial systems, protected records and printing.
Sales	6	Portable laptops, staff Wi-Fi, online CRM and presentation/collaboration tools.
Total	**20**	Four departments sharing office infrastructure with segmented access.

Business requirements

- Continuous access to Microsoft 365 collaboration services and reputable business internet connectivity.
- Least-privilege access to confidential HR and Finance information; isolated guest Wi-Fi.
- Internal development and staging capabilities without exposing unnecessary services to the public internet.
- Recoverable records, centrally managed endpoints, visible asset ownership and documented changes.

02 / Enterprise hardware inventory

Quantities below provide one primary endpoint per employee and a small amount of spare capacity. Asset IDs are proposed labeling ranges; the list is not a purchase order.

Asset ID	Hardware / target specification	Qty	Allocation and justification
HW-001-009	Desktop PCs; Core i5/Ryzen 5, 16 GB RAM, 512 GB SSD	9	HR (4), Finance (5). Fixed desks; standard business workload.
HW-010-020	Laptops; 16 GB RAM/512 GB SSD (IT models 32 GB preferred)	11	IT (5), Sales (6). Mobility, presentations, remote support.
HW-021-042	24-inch business monitors, FHD/IPS	22	One per employee plus two operational spares.
HW-043-053	USB-C laptop docking stations	11	IT/Sales desk connectivity and wired network access.
HW-054	Tower server; 8-core CPU, 64 GB ECC, 2 x 1 TB SSD RAID 1	1	IT. Ubuntu services, internal tooling and staging workloads.
HW-055	4-bay NAS; 4 x 8 TB NAS drives, RAID 6	1	IT. Approx. 16 TB usable before formatting/overhead; file and backup target.
HW-056	Dual-WAN gigabit business router	1	IT. Primary/secondary ISP failover and WAN monitoring.
HW-057	Dedicated firewall with VLAN gateway capability	1	IT. Filtering, network segmentation, VPN and security logs.
HW-058	48-port managed PoE+ gigabit switch; 10 Gb uplinks preferred	1	IT. Connects endpoints, access points and shared services; headroom.
HW-059-060	Wi-Fi 6 business access points; VLAN-aware	2	IT. Staff and isolated guest wireless coverage after site survey.
HW-061-062	Network multifunction laser printers	2	Shared. Controlled access for HR, Finance and other teams.
HW-063-064	Rack/tower UPS, 1500-2200 VA class	2	IT. Separate protected loads for network stack and server/NAS.
HW-065-066	Encrypted USB external backup drives, 8 TB each	2	IT. Rotating offline copies; not continuously connected.
HW-067	Lockable network rack with PDU, cable management	1	IT. Secure, ventilated housing for core equipment.
HW-068-087	Keyboard/mouse sets	20	All employees; consistent and replaceable peripherals.

Rationale. Desktops suit fixed HR and Finance work; laptops suit IT and Sales. The server, NAS, offline backups and UPS support recovery. A managed 48-port switch and two proposed APs allow expansion, subject to a site survey.

03 / Enterprise software inventory

Versions refer to the intended deployment family or support channel, not products already installed. The IT lead must confirm current supported builds and license entitlements before purchase.

Software	Version / channel	License / quantity	Purpose
Windows 11 Pro	Supported 64-bit release	20 OEM/commercial OS licenses	Standard managed endpoints; BitLocker, business policies and domain/cloud join.
Ubuntu Server	26.04 LTS, patched	Open source; support optional	Internal application tooling, monitoring and development/staging services. [R06]
Microsoft Office (Microsoft 365 Apps)	Current Channel	20 Business Premium user subscriptions	Word, Excel, PowerPoint, Outlook and collaboration; included in selected plan. [R04]
VS Code	Current stable	Free product license; review terms	Development, scripting, structured configuration editing; primarily IT (5).
Git	Current stable	GPL open source	Local source control, revision history and change management; IT (5).
GitHub Desktop	Current stable	Open-source desktop application	Simplified commits, branches and pushes for the IT team.
VirtualBox	7.x current stable	GPLv3 base package; extensions separate	Isolated test VMs on IT laptops, not the production server. [R07]
Google Chrome	Current stable	No-cost proprietary distribution	Web access, SaaS tools and compatibility testing; all 20 users.
Microsoft Defender	Current security definitions	Defender AV in Windows; Defender for Business via Premium	Endpoint malware protection, alerts and managed response. [R05]
AnyDesk	Current business release	Paid commercial outbound seats (initially 2)	Authorized, attended remote helpdesk sessions; restrict unattended access. [R08]
7-Zip	Current stable	Open-source/freeware (see package license)	Archive and extract approved compressed files; all endpoints.

Software	Version / channel	License / quantity	Purpose
Microsoft 365 Business Premium	Current cloud service	20 business subscriptions	Entra ID P1, Intune and Defender for Business for centrally managed identities/devices. [R05]

Licensing. Business Premium provides Office apps, Intune, Entra ID P1 and Defender for Business [R04][R05]. AnyDesk requires commercial licenses [R08]. VirtualBox extensions are licensed separately from the GPLv3 base package [R07].

04 / Enterprise network inventory

Network quantities are planned for one floor and include ISP-supplied hardware plus structured cabling. Items already listed in Part 2 are cross-referenced and must not be counted twice in procurement.

Network ID	Equipment	Qty	Technical justification
NET-01-02	ISP optical modem/ONT	2	Provider-supplied units; separate primary and backup circuits.
NET-03	Dual-WAN business router	1	Health checks, WAN route selection/failover (same unit as HW-056).
NET-04	Dedicated firewall	1	Inter-VLAN filtering, NAT/security enforcement (HW-057).
NET-05	48-port managed PoE+ switch	1	Tagged trunks, endpoint ports and PoE for APs (HW-058).
NET-06-07	Wi-Fi 6 access points	2	Staff and guest SSIDs with VLAN assignment (HW-059-060).
NET-08	48-port CAT6 patch panel	1	Structured terminations and labeled patch field.
NET-09-11	CAT6 solid-core bulk cable, 305 m boxes	3	Provisional allowance for 32 certified permanent links; survey needed.
NET-12-43	CAT6 keystone jacks / wall outlets	32	Labeled user, printer, server, AP and spare locations.
NET-44-67	RJ45 plugs for approved patch leads	24	Spare field connectors; permanent links terminate to jacks/panels.
NET-68-115	Factory-tested CAT6 patch cords	48	Rack/device patching and replacements.
NET-116	Cable labels and link-testing service	1 lot	Certify and document every permanent run.

Network ID	Equipment	Qty	Technical justification
NET-117	Rack, PDU and cable managers	1	Shared with HW-067; one procurement line, not duplicated.

Cabling design. Reserve 32 data outlets: 20 employee positions, two printers, one server, one NAS, two AP locations and six spare points. Use solid-core CAT6 for permanent runs terminated at jacks and the patch panel; use factory-tested patch cords for device connections. The three 305 m boxes are a planning allowance only and must be revised after measuring the actual floor.

05 / Enterprise network diagram

Figure 1 shows the logical enterprise topology, including two independent internet circuits, modem/ONT units, a dual-WAN router, a stateful firewall, a managed access switch, departmental endpoints and shared services. This design intentionally keeps internal VLAN routing and access policy on the firewall.

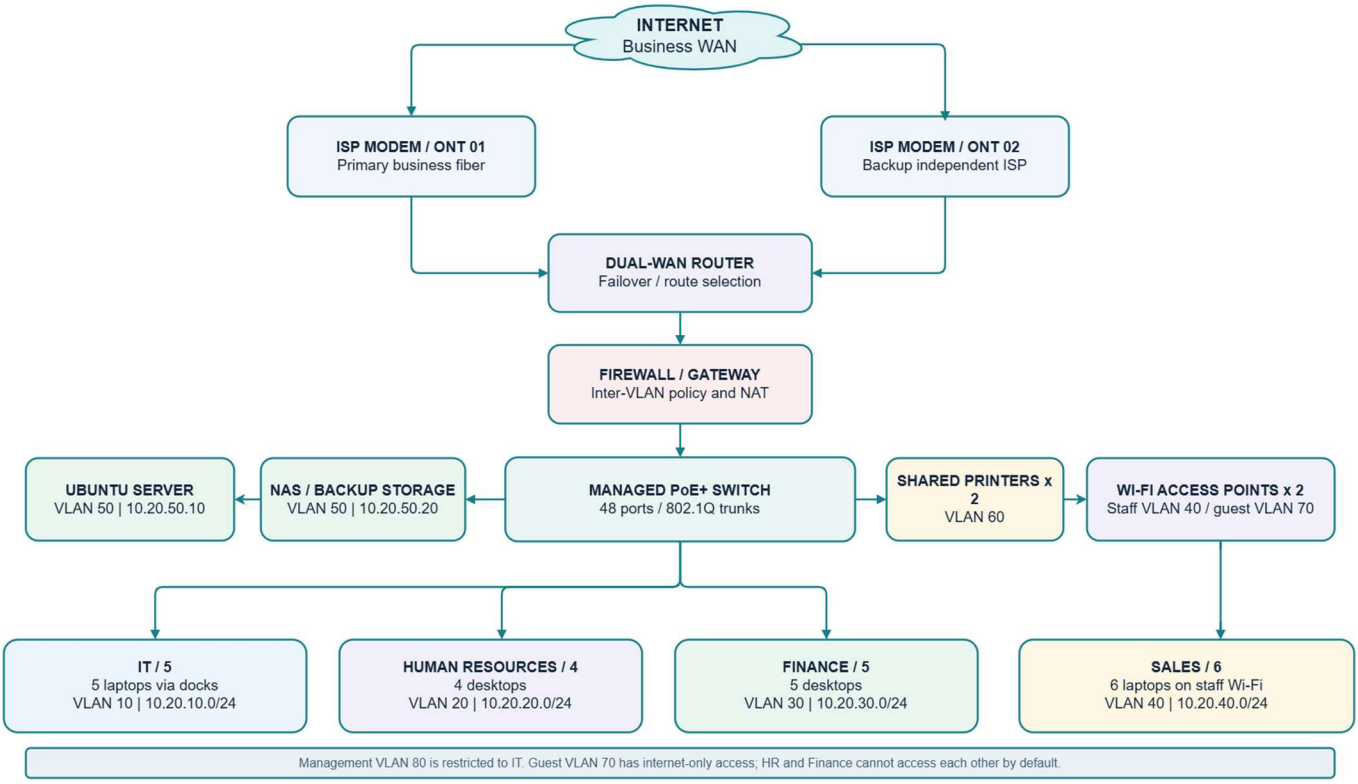


Figure 1. Proposed logical topology. A full-size PDF/PNG and an editable diagrams.net file are in the diagrams folder.

Logical addressing and segmentation

VLAN	Zone	IPv4 subnet	Primary use / rule
10	Information Technology	10.20.10.0/24	5 IT laptops; authorized admin workflows
20	Human Resources	10.20.20.0/24	4 desktops; HR data isolated from Finance

VLAN	Zone	IPv4 subnet	Primary use / rule
30	Finance	10.20.30.0/24	5 desktops; accounting-only services
40	Sales / staff Wi-Fi	10.20.40.0/24	6 laptops; business SaaS access
50	Server / NAS	10.20.50.0/24	Server .10 and NAS .20; approved application flows
60	Printers	10.20.60.0/24	Two shared printers; source-restricted printing
70	Guest Wi-Fi	10.20.70.0/24	Internet only; no local LAN routes
80	Management	10.20.80.0/24	Switch/AP/firewall admin; restricted IT access

Baseline rule set. Deny direct cross-department connections by default. Allow only documented destinations/ports for business services, permit approved print traffic to VLAN 60, deny printers initiating sessions toward clients, and restrict server/NAS access by application role. Guest VLAN 70 may reach the internet only. VLAN 80 is administrator-only and must not be used for ordinary browsing. IT privileges require separate administrator accounts and MFA wherever supported.

Addressing and wireless. The firewall provides VLAN gateways, DHCP scopes and DHCP reservations. Reserve 10.20.50.10 for the Ubuntu host and 10.20.50.20 for the NAS; reserve a small static range for printers and management devices. Use a staff SSID mapped to VLAN 40 for Sales laptops and a separate guest SSID mapped to VLAN 70. Pilot WPA2/WPA3-Enterprise authentication where client devices and identity infrastructure support it; otherwise use managed WPA3 configuration with per-network credentials while planning the enterprise transition.

06 / System administration roles

In a 20-person startup these responsibilities may be distributed among the five existing IT employees rather than requiring four separate full-time hires. The roles remain distinct for accountability and escalation.

1. Helpdesk Technician

Responsibilities. Handles first-line incidents, device provisioning, account requests, ticket triage, software installation and escalation. Provides user-oriented support while recording recurring failures and their resolution.

Core skills. Windows troubleshooting, clear communication, ticket classification, remote assistance, basic security awareness.

Typical tools. Ticketing system, Windows Event Viewer, Microsoft Intune, approved AnyDesk sessions, knowledge base.

Relevant certifications. CompTIA A+; Microsoft 365 Fundamentals is useful background.

2. Network Administrator

Responsibilities. Designs and maintains switches, VLANs, routing, ISP connectivity, Wi-Fi, firewall policies and monitoring. Investigates outages and validates segmentation rules before change approval.

Core skills. TCP/IP, subnetting, switching, wireless design, firewalls, packet capture, change control.

Typical tools. Managed-switch console, firewall portal, ping/traceroute, Wireshark, SNMP monitoring.

Relevant certifications. CompTIA Network+; Cisco CCNA.

3. Linux System Administrator

Responsibilities. Installs and hardens Ubuntu Server; manages users, services, logs, patches, backups and recovery tests. Supports internal development and staging workloads.

Core skills. Bash scripting, SSH, systemd, permissions, storage, package management, monitoring and incident response.

Typical tools. SSH, journalctl, systemctl, apt, ufw, rsync and monitoring/backup agents.

Relevant certifications. Linux Foundation LFCS; Red Hat RHCSA for a Red Hat-oriented track. [R10][R11]

4. Cloud Administrator

Responsibilities. Manages Microsoft 365 tenant identity, licensing, endpoint-management policy, cloud storage, access review, backup integrations and cloud costs. Coordinates governance and secure hybrid access.

Core skills. Identity/access management, MFA, Entra ID, Intune, cloud networking, policy-as-code basics and cost controls.

Typical tools. Microsoft 365 admin center, Entra admin center, Intune, Azure portal, cloud monitoring.

Relevant certifications. Microsoft Azure Administrator Associate (AZ-104); AWS Certified CloudOps Engineer - Associate for AWS specialization. [R09][R12]

How the four roles work together

The helpdesk receives an issue, documents its business impact and performs first-line troubleshooting. If connectivity or Wi-Fi is involved, the network administrator checks segmentation, WAN health and firewall policy. If a local application or server is failing, the Linux administrator investigates service state, system logs and recovery options. For Microsoft 365 sign-in, device enrollment or a cloud-hosted dependency, the cloud administrator checks identities, access and service health. They coordinate through shared incident records, approved changes and escalation rules so a fix does not unintentionally weaken security elsewhere.

07 / Infrastructure recommendations

Internet provider and resilience

Seek independent site-qualified quotes from **Globe Business** and **PLDT Enterprise** for business fiber access [R13][R14]. Target 300-500 Mb/s for the primary service and 100-200 Mb/s for a secondary circuit on a different provider and, where practical, a different last-mile route. Compare guaranteed upload performance, actual serviceability, support response, reliability, IP requirements and termination terms; advertised "up to" rates are not guarantees. Verify automatic failover with a planned outage test.

Server and storage specification

Use one business-grade 8-core tower server with 64 GB ECC memory, mirrored enterprise SSDs (2 x 1 TB), protected power, remote management and room for additional storage. Install a supported Ubuntu Server 26.04 LTS build [R06]; host only internal development/staging tools and monitoring with SSH limited to authorized IT staff. Use one RAID-6 NAS with 4 x 8 TB NAS disks as local shared/backup storage; RAID is availability protection and **not** a substitute for backup.

Backup and recovery strategy

- Follow a 3-2-1 approach: primary data plus two backups, stored on at least two media types, with one copy offsite/offline or immutable [R03].
- Run nightly incremental backups and weekly full backups for local server/NAS data; protect important Microsoft 365 data with a separately evaluated cloud backup service.
- Use two encrypted external drives in a documented rotation; disconnect offline media after the backup and keep one in secured offsite custody.
- Proposed recovery targets: RPO within 24 hours for critical documents; RTO within 8 business hours for priority services, subject to testing and executive acceptance.
- Test a sample restore monthly and complete a full documented recovery rehearsal quarterly; retain daily, weekly and monthly backup sets according to approved data policy.

Security baseline, antivirus and passwords

- Mandate MFA for Microsoft 365, remote administration and privileged identities; prefer phishing-resistant methods for administrators. Apply least privilege and separate standard/admin identities [R01][R02].
- Enroll all Windows 11 Pro endpoints in Intune; enable BitLocker, host firewalls, Microsoft Defender for Business, screen locks and application updates [R05].
- Require long unique passphrases (15+ characters for password-only logins), block common/compromised passwords, encourage a password manager and do not force routine password changes without a compromise signal.
- Apply critical security patches rapidly by risk and service impact; document patch windows, centralize logging and use alerting for failed backups, unusual sign-ins and security detections.

- Use approved, authenticated remote-support sessions only; disable unattended access unless individually justified and governed. Maintain an incident-response contact list and access revocation checklist.

Expansion and phased implementation

Phase	Activities	Acceptance evidence
0 - Approve	Floor survey, vendor quotes, procurement review, license and risk sign-off.	Signed inventory/specification and implementation schedule.
1 - Build	Deploy cabling, dual ISP, router, firewall, switch, APs, rack and UPS.	Link-test records; WAN failover test; coverage survey.
2 - Secure	Configure 8 VLANs, firewall rules, tenant identities, Intune and endpoint policies.	Access-control tests; MFA/BitLocker/Defender compliance.
3 - Validate	Provision Ubuntu host, NAS, backups, printers, user devices and documentation.	Restore test, endpoint inventory, user acceptance sign-off.
4 - Grow	Plan 30 users in 12-18 months; use spare switch ports and scalable licenses.	Updated capacity forecast and budget; assess second firewall/server.

Priority risks. A single firewall, switch and server remain points of failure. Dual ISP protects only against upstream outages, not loss of local power or network hardware. UPS protects short interruptions, not extended outages. Document risk acceptance initially, test recovery, and evaluate redundant core devices as headcount and business criticality increase.

08 / Personal reflection

This project helped me understand that system administration begins long before computers, servers, or network devices are installed. I learned how to translate a company's business requirements into a practical infrastructure plan. In the case of ABC Startup Solutions, the four departments have different needs even though they share one office. IT requires reliable development and administrative tools, Human Resources handles confidential employee information, Finance processes sensitive financial records, and Sales depends on fast communication and mobility. Recognizing these differences helped me understand why the same equipment or network permissions should not automatically be assigned to everyone.

The most challenging part of the project was creating a network design that remained simple enough for a small startup while still addressing security and future growth. At first, it seemed sufficient to connect all devices to one switch and wireless network. However, planning the VLANs, firewall rules, shared printer access, and separate guest Wi-Fi showed me how easily an unsegmented network could expose one department's information to another. I also had to balance the equipment quantities against real operational needs, such as purchasing enough laptops for mobile employees, choosing a switch with spare ports, and including backup power and storage instead of focusing only on computers.

I also learned the value of documentation in preventing confusion during deployment and maintenance. An inventory makes it easier to track assets and licenses, while a network topology shows how devices communicate. Planning a backup schedule, recovery targets, password policy, and administrator responsibilities before deployment gives the IT team a clearer basis for testing and responding to failures. Using reputable technical sources also reminded me that recommendations should be supported by established practices rather than personal assumptions.

As a fourth-year Information Technology student beginning to pursue Data Analytics, I can see how this activity connects to my broader professional goals. Reliable infrastructure, controlled access, and accurate documentation are necessary for protecting the data that organizations later analyze. This project strengthened my technical reasoning and showed me how to communicate decisions in a format that a manager could review. Moving forward, I want to develop these skills through practical configuration exercises and improve my ability to design systems that are secure, maintainable, and appropriate for the people using them.

References and source notes

The references below support general security, software licensing, support lifecycles, certification pathways and ISP availability research. Hardware quantities, VLAN addressing and recovery targets are scenario-based professional recommendations, not sourced purchasing quotes. Links were checked on August 20, 2026.

[R01] NIST: CSF 2.0 for SMBs	[R02] NIST: Cybersecurity Basics
[R03] CISA: 3-2-1 Data Backup	**[R04]** Microsoft 365 Apps
[R05] Microsoft 365 Security FAQ	**[R06]** Ubuntu LTS Lifecycle
[R07] VirtualBox Licensing	**[R08]** AnyDesk Commercial Use
[R09] Azure Administrator AZ-104	**[R10]** Linux Foundation LFCS
[R11] Red Hat RHCSA	**[R12]** AWS CloudOps Engineer
[R13] Globe Business Fiber	**[R14]** PLDT Business Connectivity